

Predicting the student's perceptions of multi-domain environmental factors in a Norwegian school building: Machine learning approach

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ABSTRACT

Poor Indoor Environmental Quality (IEQ) in schools significantly impacts students' well-being, learning capabilities, and health. Perceived dissatisfaction rates (PD%) among students often remain high, even when indoor environmental variables appear well-controlled. This study aims to predict perceived dissatisfaction rates (PD%) across multi-domain environmental factors—thermal, acoustic, visual, and indoor air quality (IAQ)—using machine learning (ML) models. The research integrates sensor-based environmental measurements, outdoor weather data, building parameters, and 1437 student survey responses collected from three classrooms in a Norwegian school across multiple seasons. Statistical tests were used to pre-select relevant input variables, followed by the development and evaluation of multiple ML algorithms. Among the tested ML models, Random Forest (RF) demonstrated the highest predictive accuracy for PD%, outperforming multi-linear regression (MLR) and decision trees (DT), with R^2 values up to 0.91 for overall IEQ dissatisfaction (PDIEQ%). SHAP analysis revealed key predictors: CO₂ levels, VOCs, humidity, temperature, solar radiation, and room window orientation. IAQ, thermal comfort, and acoustic environment were the most influential factors affecting students' perceived well-being. Despite limitations as implementation in building level scale, the study demonstrates the feasibility of deploying predictive ML models under real-world constraints for improving IEQ monitoring system. The findings support practical strategies for adaptive indoor environmental management, particularly in educational settings, and provide a replicable framework for future research. Future research can expand to other climates, buildings, measurements, occupant levels, and ML training optimization.

1. Introduction

Schools are the second place after homes, where children spend the most time [1–3]. Poor Indoor Environmental Quality (IEQ) in school buildings can lead to various symptoms, illnesses, and health problems for children [4,5]. A positive correlation has been found, indicating that pupils are more vulnerable than adults due to their weaker immune systems, [6–9] making them more susceptible to multi-domain

environmental factors such as thermal, acoustic, visual, and indoor air quality (IAQ) [6,10]. As occupants' well-being in buildings is significantly influenced by surrounding IEQ conditions [11], it is crucial to examine how environmental factors play a significant role in maintaining students' well-being. Ensuring optimal IEQ is essential for providing students and school staff with a comfortable, productive, and safe indoor environment [12].

IEQ in classrooms is commonly measured using sensors [13]. IEQ in

Abbreviations: ML, Machine Learning; MLR, Multi Linear Regression; DT, Decision Tree; RF, Random Forest; PD%, Percentage of Dissatisfied; IAQ, Indoor Air Quality; IEQ, Indoor Environment Quality; TE, Thermal Environment; AE, Acoustic Environment; VE, Visual Environment; VOCs, Volatile Organic Compounds; CO₂, Carbon dioxide; RH, Relative Humidity.

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schools, especially in Norway, is mostly provided with mechanical ventilation systems and artificial lighting. However, indoor contaminants such as elevated CO₂ levels often slightly exceed recommended limits [14,15], while VOCs released by indoor materials and furniture [16,17], as well as pollutants and fine-particles [18,19] remain concerns. Building parameters such as indoor characteristic also affect occupants' satisfaction [20,21]. Indoor setting condition correlated with space heavily influences overall satisfaction [22]. Despite maintaining IEQ within established standards, occupants' satisfaction is not always highly achieved [23]. For example, the maximum allowable dissatisfaction rate for thermal comfort is 25% of the occupants in a room should be dissatisfied [24]. Therefore, a method is needed to measure and predict the perceived dissatisfaction (PD) rate (PD%) of multi-domain environmental factors.

User-feedback is a promising method for collecting environmental comfort perceptions across various factors [11,25]. Providing user feedback allows students to respond in real-time about their satisfaction/dissatisfaction, offering valuable insights for building operation purposes. By collecting and analysing IEQ data alongside user feedback, satisfaction/dissatisfaction responses can be observed to better understand occupant experiences [26,27].

Environmental dissatisfaction approaches, such as PMV-PPD for thermal comfort factors, are commonly available and inspired the way to understand how perceived indoor environment by occupants, especially thermal comfort. However, thermal comfort models but remain insufficiently accurate for prediction purpose in specific place [28]. Besides, machine learning (ML) models are widely used in IEQ research [29–32], offering promising predictive applications for future advancements [33]. Support vector machine (SVM) [22], multivariate linear regression [34–36] and non-linear regression algorithms [37–40] are commonly employed to classify and estimate overall IEQ satisfaction. Statistical learning methods like ML, such as SVM [41–43], classification trees [41,42] and random forest [41–45] have been applied to thermal comfort predictions, while artificial neural network (ANN) model has been utilized for predicting thermal, visual, and acoustic environments [46]. Moreover, ML algorithms have gained significant attention for predicting overall IEQ satisfaction [44]. Leveraging these advancements, ML provides a robust tool for predicting students' perceptions of multi-domain environmental factors using various input parameters. The ML models developed in this study introduce a novel approach for optimizing indoor environments especially in school buildings.

Table 1, Table 2

Although ML methods have been applied to predict specific IEQ domains, their application for predicting dissatisfaction rates (PD%) across multi-domain environmental factors simultaneously remains less developed, especially in real application. This study introduces ML approach of real implementation in building-scale by collecting occupant feedback to understand preference of dissatisfaction rates (PD%) based on variations of environmental variables dependently. Environmental variables and building parameters are combined with student dissatisfaction surveys to develop ML models capable of predicting the PD% for thermal, acoustic, visual, and indoor air quality (IAQ) domains. The objective of this study is to predict the PD% of multi-domain environmental factors using ML methods based on building characteristics and environmental variables. Three research questions are addressed: 1) How accurately can ML methods predict the PD% of multi-domain environmental factors 2) Which input parameters significantly contribute to ML predictions of multi-domain environmental factors 3) How environmental factors influence the PD% of well-being, assumed to represent the overall IEQ dissatisfaction rate. This study is limited to a specific school, with the scope focusing on predicting PD% at the school level and their relationship to student PD%. Certain assumptions are made to generalize the findings, suggesting that this study may represent similar cases in comparable settings.

Table 1

Summary of research related to multi-domain environmental domains and occupants' perceptions.

Ref.	Location	IEQ Domains	Modelling Approach	sample sizes	Approaches
[37]	Office	Multi-domain (thermal, IAQ, acoustic), Overall IEQ	multivariate non-linear regression	422	Field
[36]	Office	Multi-domain	multivariate	68	Experiment
[34]	University	Multi-domain	linear and non-linear	500	
[38]	School	(thermal, IAQ, visual, acoustic), overall IEQ	regression	224	
[35]	School	Multi-domain (thermal, visual, acoustic), overall IEQ	multivariate linear regression	115	Field
[39]	Office	Thermal, Visual	linear mixed model	75	
[40]	School	Thermal, Visual, Acoustic, IAQ	Regression Analysis	55	
[46]	University	Thermal, Acoustic, Visual	ML (ANN)	35	Experiment
[45]	Residential	Thermal comfort	ML (Random Forest)	7	
[41]	Office	Thermal comfort	ML (RF, SVM, GradBoost)	34	
[43]	University	Thermal comfort	ML (SVM, DT, RF)	12	Experiment
[42]	Office	Thermal comfort	ML (SVM, DT, RF)	14	
[22]	Office	Overall IEQ	PCA, ML (SVM, RF)	663	
[44]	Office	Multi-domain (thermal, IAQ, visual, acoustic), overall IEQ	ML (RF, XGBoost)	56	Experiment (dataset-I)

Table 2

Summary of varied building interior parameters of each classroom.

Classroom Number:	Window Area/Floor Area (m ² /m ²)		Room Location	Room Size (m ²)	Ceiling Height (m ²)
	East	West			
11	0	0.22	1 st floor	61	2.8
23	0	0.24	2 nd floor	57	2.8
27	0.22	0	2 nd floor	61	2.9

2. Research methods

The research methods in this study combined field measurements of indoor and outdoor environmental parameters, building parameters, students' survey, and machine learning (ML) methods, which were used to model and predict perceived dissatisfaction rate (PD%) of multi-domain environmental factors. By applying statistical tests and ML models with interpretative analysis, the study identified the most significant input parameters influencing PD% and evaluated the predictive performance of different ML algorithms. This methodological approach enables a data-driven analysis of IEQ, ensuring localized prediction of

dissatisfaction trends.

2.1. Buildings and environmental conditions

The environmental conditions surrounding students play a crucial role in identifying their dissatisfaction levels as occupants of school building. In this study, field data on building parameters and environmental variables were collected and combined into a single dataset for analysis.

2.1.1. Building parameters

In this study, we used data from a secondary school in Oslo municipality – Norway, a cold climate country, 50 years old building structure with dry humidity climate, with recent significant renovation on interior and building system in 2008. Three classrooms were included based on availability and willingness of teachers to participate. Certain building parameters were documented and described in this study. The classrooms are situated on different floors and sides of the building, and room indoor characteristics are shown in Fig. 1. Each room has one external façade; with walls and windows in direct contact with the outdoors. The windows, installed as a double-glazing construction with an external film coating, are almost two decades old and face either the east (hill) or west (sea-view sides). The ventilation system in this school relies on an

air-handling unit (AHU), centrally located on the rooftop. AHU supplies fresh air to all rooms with the same amount as designed capacity ($\pm 1400 \text{ m}^3/\text{h}$) although is not recorded. AHU applied setpoint on CO_2 levels in accordance with local recommendation ($<1000 \text{ ppm}$) [15]. The modulation of the AHU is centrally regulated, ensuring that air flow variations are uniform across all rooms. The heating system in all rooms uses hydronic radiators to maintain a set temperature of 21.0°C during study hours and 19.0°C outside study hours and on weekends. Building parameters provide insight into the variations in indoor environmental conditions between rooms on dissatisfaction of environmental factors.

2.1.2. Environmental variables

The environmental variables in this study include both indoor and outdoor environmental data. Indoor sensor data in this study is important to understand objective IEQ, as this device is commonly installed in the classrooms. Integrated sensors connected to the AHU (used as control inputs) and new free-standing compact low-cost sensor devices (*Airthings®*) were installed in each classroom, as detailed in Table 3. Air temperature as dry bulb, relative humidity, and barometric air pressure measurement are provided to indicate indoor thermal condition. Carbon dioxide (CO_2) as well as Volatile Organic Compounds (VOCs) are provided to indicate air quality; and relative light intensity are provided to represents the proportion of detected light relative to the sensor's

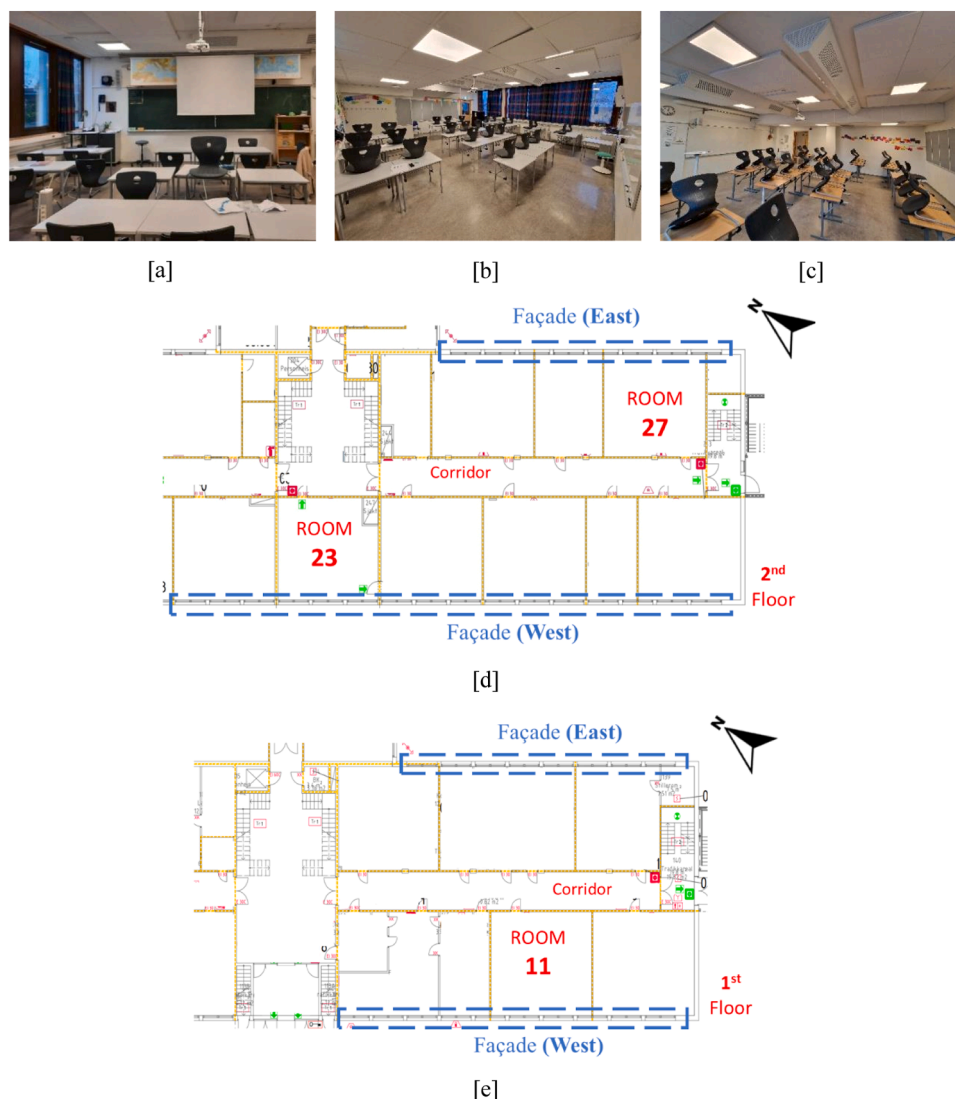


Fig. 1. Documentation of classroom conditions: (a) Room 11, (b) Room 23, (c) Room 27, (d) First Floor Plan, and (e) Second Floor Plan.

Table 3

Characteristics of the capabilities of measuring instruments installed in each classroom.

Devices	Parameters	Unit	Accuracy	Resolution	Operating range
Sensors by AHU	CO ₂	ppm	30 (±3%) ± 0.1	1 0.1	400 – 5000 4–30
Free-Standing Sensors	Air Temperature	°C	± 0.1	0.1	4–30
	CO ₂	ppm	30 (±3%) ± 0.1	1 0.1	400 – 5000 4–30
	Air Temperature	°C	± 0.1	0.1	4–30
	Relative Humidity (RH)	%	± 1	1	0 - 80
	VOCs	ppb	5 (± 3%)	1	400 – 5000
	Barometric Air Pressure	hPa	± 0.15	0.1	500–1100
	(Relative) light intensity	%	–	1	1–100

maximum capacity. Both sensor types were mounted side-by-side on walls at a height of less than 1.5 m above the floor, in accordance with the relevant installation standard [47]. As newly purchased sensors, these instruments are assumed operating with reliable accuracy, measuring continuously 24 hours a day from beginning until end of this field observation (March 2023 to April 2024).

Outdoor environmental data in this study data through online [48] provided by local closest weather station in municipality at 1-hour intervals, located approximately 1.2 km from the school and measured available parameters: (1) outdoor air temperature (°C); (2) outdoor relative humidity / RH (%); (3) outdoor wind speed (m/s); and (4) outdoor precipitation (mm); (5) barometric air pressure (hPa); (6) solar horizontal radiation (W/m²). Specific humidity (g/kg dry air) and air enthalpy (kJ/kg) were calculated as additional parameters using provided data (air temperature, RH, and air barometric pressure) based-on psychrometric chart [49]. The ventilation system operated in heating mode, resulting in indoor specific humidity being consistently higher than outdoor levels. Indoor moisture gain (Δ_g /kg dry air) was calculated as the difference between the specific humidity of indoor and outdoor air. To ensure compatibility with indoor environmental data, outdoor environmental data were interpolated linearly to 10-minute timestep.

2.2. Students' survey

The students' survey served as a direct method for collecting user feedback. In this study, the survey gathered students' satisfaction level responses regarding multi-domain environmental factors, including IAQ, thermal, acoustic, and visual environmental-related questions. Secondary school students are 13–15 years old, participated in the survey voluntarily and repeatedly. Teachers were also present to assist students if they encountered any difficulties. The questionnaire was completed by all students in each classroom both in the morning and afternoon. The survey was conducted repeatedly over multiple days. Longitudinal user-feedback by diverse respondents may bring understanding on human comfort in dynamic environments [45].

The survey consisted of four questions addressing multi-domain environmental factors, with four-scale preference options. All questions were phrased as "How is ... now?" and followed by an overall feeling question at the end. This approach was used to determine the PD% of environmental factors for: "Room Air" – as IAQ (PD_{IAQ} %), "Room Temperature" – as Thermal Environment (PD_{TE} %), "Room Noise" – as Acoustic Environment (PD_{AE} %), "Room Light" – as Visual Environment (PD_{VE} %). Last question "How do you Feel today?" for overall well-being perception (PD_{WB} %), effecting the percentage of students feeling well. The answer scale included the options: *Very Dissatisfied*, *Dissatisfied*, *Satisfied*, and *Very Satisfied*. Icon to represent 'neutral' option is not

shown but recorded as default value if students skip or do not select any satisfactory option in respective question. Neutral responses were interpreted as potentially indicating either a lack of engagement or difficulty in understanding the survey questions, and this possibility was considered during data interpretation. PD% of all environmental factors were calculated individually for each batch (morning or midday) using Eq. (1).

Teachers guided once in the beginning the students filling the survey. Teachers reminded students to finish surveys completed online in two time batches, morning and midday. To minimize response bias, teachers informed nothing to students about the research purpose. Morning batches were assumed to represent a 1-hour occupancy hour, while midday batches represented a 5-hour occupancy hour in the same classroom. These time-based responses enabled the evaluation of occupants' adaptation to the surrounding environment.

The surveys were administered online using unique internet links assigned differently to each classroom, as shown in Fig. 2. The survey was designed to be user-friendly, requiring students spontaneously to select their responses and click 'Next' to complete it, with no personal identification required. Participation was entirely voluntary, and students could choose to skip questions or withdraw without any consequences. Responses were collected anonymously, with no names, gender, or identifiable data recorded, ensuring compliance with Norwegian privacy regulations. Teachers were present to assist if any students encountered difficulties. Once Students arrive at school, normally students store their outdoor jackets in lockers before entering classrooms. This public school do not wear uniforms by chance, assuming students may self-regulate their clothing style based on personal comfort. Besides, students' activity levels were assumed to be uniform (reading & sitting), resulting in relatively similar metabolic rates throughout the survey period. In this study, activity level and clothing style were not included coincidentally in survey to prioritize response efficiency and ensure straightforward participation.

$$PD_{\text{factors}}(\%) = \frac{\text{Very Dissatisfied}_{\text{count}} + \text{Dissatisfied}_{\text{count}}}{\text{Total Responses}_{\text{count}}} \quad (1)$$

2.3. Statistical tests & machine learning models

Statistical tests were conducted in this section to pre-select significant input parameters, ensuring that ML models are efficiently developed and capable of predicting PD% with validated accuracy. Subsequently, *SHapley Additive exPlanations* (SHAP) analysis was applied to interpret the contribution of individual input parameters to the model predictions.

Two primary statistical tests, ANOVA one-way and Chi-squared, were employed to explore the relationship between building and environmental variables and dissatisfaction levels. The ANOVA one-way test was used to identify significant variations in dissatisfaction levels across

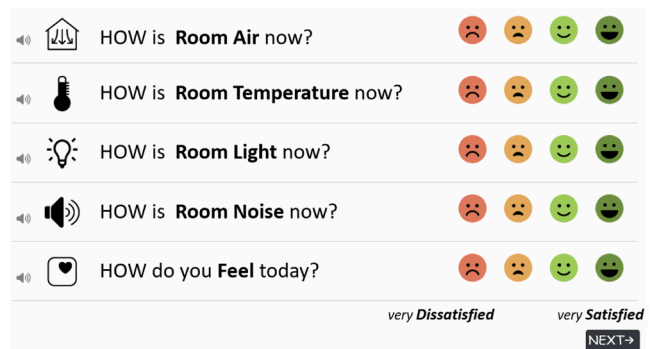


Fig. 2. Dashboard displays of students' survey (translated into English from the origin language: Norwegian).

different building and environmental variables, enabling the identification of parameters that influence perceived dissatisfaction. For example, this test helps determine whether measurements significantly affect satisfaction/dissatisfaction distribution. The Chi-squared test assessed the association between dissatisfaction levels and categorized environmental variables, such as high versus low humidity or temperature ranges. Parameters with p-values < 0.05 in either test were considered statistically significant, indicating a rejection of the null hypothesis (no correlation) with 95% confidence. These tests were critical for pre-selecting significant parameters, ensuring efficient development ML models.

PD% of each environmental factor as calculated with Eq. (2), offering a clearer representation of the degree of satisfaction or dissatisfaction across the entire occupant group. This allows for a more nuanced understanding of how environmental factors influence environmental dissatisfaction trends and practical. ML algorithms were proposed to predict the PD% of environmental factors in this study, including multi-linear regression (MLR), Decision Tree (DT), and an ensemble method (Random Forest/RF). MLR derives linearly the analysis of pre-selected input parameters as variables ($x_1, x_2, \dots, x_p$) with trained coefficients ($b_0, b_1, \dots, b_p$), and is used to predict PD% of environmental factors as shown in Eq. (2). By leveraging the pre-selected input parameters from statistical tests, regression-based ML algorithms can be efficiently developed to predict the PD% of environmental factors. DTs are widely used as predictive models or decision-support tools due to their interpretability [50,51], including applications in predicting satisfaction and IAQ [41,42,52]. However, a fully developed DT may be prone to overfitting, as splits are repeated on subsets until the terminal nodes contain very few target values, potentially resulting in overly complex trees [53]. RF, another promising ML algorithm for predicting satisfaction [41–44]. RF is a combined random-trees predictor; each tree is based on the values of a random vector generated independently and with the same distribution for all trees in the forest [53]. This study anticipates that RF can address common overfitting issues often observed in DT models.

Cross-validation method was employed to ensure the reliability and robustness of the ML models. The dataset was randomly split into an 80% training set and a 20% test set. *GridSearchCV* with default 5-fold cross validation was used for hyperparameter tuning to optimize model performance for simplicity in modelling. all hyperparameter combinations are listed in Table 4. The evaluation of the models used Root Mean Squared Error (RMSE) as performance metrics and was presented using R-squared (R^2). RMSE was employed to measure prediction error magnitude, while R^2 assessed the proportion of variance explained by the models. This validation framework ensured that the selected ML models were not only well-trained but also generalizable to unseen data, reducing the risk of overfitting and improving predictive accuracy.

Machine learning (ML) are often considered difficult to explain, as typically are assumed to function as black boxes [54]. SHAP analysis ensures interpretability by explaining how each input parameter contributes to the model's predictions, offering actionable insights for optimizing indoor environments. SHAP is commonly used to interpret complex ML models, such as RF [55,56]. In this study, SHAP analysis was applied to reveal the interactions and contributions of each pre-selected input parameter to the predicted PD% by students. The relative importance of input parameters was assessed to determine their influence on the ML model's predictions. SHAP values provided a

quantitative measure of each feature's contribution to dissatisfaction levels, highlighting key variables that significantly affect PD% across different environmental domains. Higher SHAP values indicate stronger influences. The distribution of SHAP values further illustrated the interactions between environmental and building parameters as ML inputs.

$$PD_{pred,i} = b_0 + b_1x_1 + b_2x_2 + \dots + b_px_p \quad (2)$$

$$R^2 = 1 - \frac{\sum_{i=1}^n (PD_{real,i} - PD_{pred,i})^2}{\sum_{i=1}^n (PD_{real,i} - \overline{PD_{real}})^2} \quad (3)$$

$$RMSE = \sqrt{\frac{\sum_{i=1}^n (PD_{pred,i} - PD_{real,i})^2}{n}} \quad (4)$$

Data organization, processing, and ML fitting was done in computer. *Python programming* was used as working platform by this study with *Pandas*, *NumPy*, *psycholib*, *Matplotlib*, and *scipy* libraries as tools of data-processing, psychrometric calculation, visualization and statistic tests process. Subsequently, *scikit-learn* and *shap* libraries were used for all MLs process and SHAP analysis. *GridSearchCV* as a tool from *scikit-learn* library was used for cross-validation technique.

3. Results & discussions

The results section provides an in-depth analysis between building parameters, environmental variables and student dissatisfaction levels. The analysis results herein involved descriptive statistics, statistical tests, and ML methods to derive insights from the data and predict student dissatisfaction using environmental variables. Indoor & Outdoor Environment, and user-feedback data collection campaigns were conducted in three times campaign, in March–April 2023, November 2023, and February–March 2024, assumed this study covered all seasons (the spring and autumn semesters).

3.1. Characteristics of collected data

A survey was conducted involving approximately 80 students (mixed men and women) across three different classrooms, with data collected during four data collection periods, each lasting about ten days (March & November 2023, February & April 2024). A total of 1437 user-feedback was obtained, establishing this study as a longitudinal model. As general surveys and polls, response rate (>30%) of PD% data were used to ensure reasonable error margin (< 7.2%) at a 90% confidence level for general survey model [57]. Dissatisfaction level responses were transformed into PD% using Eq. (1), paired with environmental variables aggregated as mean values. In most cases, students tended to complete all questions, ensuring comprehensive response data. However, responses where students allowed it to be filled with default values were few. In extreme point, responses with all questions unanswered also existed. Thus, responses with neutral option excluded from the dataset to avoid confuse and maintain data integrity [58]. Besides, all data from measurements were filtered specifically for the days when questionnaire survey feedback was collected. Descriptive statistics of the measurements (indoor and outdoor weather), along with the variation in PD% for environmental factors, are presented in Table 5.

In this school building, the indoor environmental conditions are fully controlled by heating, mechanical ventilation, and indoor artificial lighting as typically exist in Nordic country such as Norway. However, they may partially depend on changes in outdoor weather conditions. Mean CO₂ levels by free standing sensors (660 ppm) marked the rooms were in well-controlled conditions but occasionally exceeded the set-point (>1000 ppm), suggesting delays in the air handling unit (AHU) response time to adjust airflow to its maximum capacity. The CO₂ and temperature readings from AHU sensors closely aligned with those from

Table 4
Hyperparameter lists for grid optimisation of ML algorithms.

Parameters	Variation	Algorithms to use
Max_depth	2,3,4,5, 6, 7,8,10, 12,15,20,25,50	DT, RF
Max_features_use	2,3,4,5, 6, 7,8,10	DT
n_estimators	5, 10, 25, 50, 75, 100, 200	RF

Table 5
Descriptive Statistic of environmental variables (indoor & outdoor) and collected responses.

Source	Variables	unit	Stats:				Dist.	Parameters
			std	Min	mean	max		
Indoor: Sensors by AHU	Air Temp.	°C	1.4	17.8	22.4	25	W	$L = 13.1, Sc = 9.9, Sh = 8.1$
	Carbon dioxide	ppm	193	380	712	1294	G	$Sh = 148, Sc = 15.7$
Indoor Data: Free-standing Sensor	Air Temp.	°C	1.0	17.0	21.0	23	W	$L = 21.0, Sc = 0.54$
	Relative humid.	%	6.1	14	26	45	G	$Sh = 6.0, Sc = 2.5$
	Air pressure	hPa	13.3	967	997	1023	N	
	Carbon dioxide	ppm	152	420	660	1541	G	$Sh = 23, Sc = 32$
	VOCs	ppb	138	46	213	883	G	$Sh = 1, Sc = 164$
	Light Intensity	%	26.4	0	48.4	123	LN	
	Air enthalpy	kJ/kg	2.9	23.72	31.4	38.64	G	$Sh = 23, Sc = 0.62$
	Spec. humidity	g/kg dry air	1.0	2.03	4.1	6.43	G	$Sh = 5.8, Sc = 0.42$
	Moisture gain	$\Delta g/kg$ dry air	0.4	0	0.6	1.54	W	$L = 0, Sc = 1.34, Sh = 2.0$
	Temperature	°C	6.0	-4.6	3.8	18.7	G	$Sh = 1.67, Sc = 5.1$
Outdoor Data	Relative humid.	%	20.3	26	69	100	N	
	Air pressure	hPa	13.0	972	1002	1028	N	
	Wind speed	m/s	1.5	0.4	3.1	7.7	W	$L = 0.20, Sc = 3.3, Sh = 2.0$
	Precipitation	mm	0.2	0	0.0	1.6	G	$Sh = 0.21, Sc = 0.66$
	Solar radiation	W/m ²	218.0	0	180.4	710	W	$L = -2, Sc = 122, Sh = 0.6$
	Air enthalpy	kJ/kg	8.0	-0.98	12.3	32.2	G	$Sh = 2.3, Sc = 5.8$
	Spec. humidity	g/kg dry air	1.2	1.26	3.4	6.47	W	$L = 1.22, Sc = 2.5, Sh = 2.0$
	PD _{IAQ}	%	21 %	11 %	52 %	93 %	W	$L = 0.06, Sc = 0.47, Sh = 2.4$
	PD _{TE}	%	20 %	6 %	46 %	86 %	N	
	PD _{VE}	%	14 %	8 %	37 %	73 %	N	
Dissatisfied Students	PD _{AE}	%	20 %	0 %	33 %	82 %	W	$L = 0, Sc = 0.4, Sh = 1.8$
	PD _{WB}	%	20 %	12 %	48 %	91 %	W	$L = 0.06, Sc = 0.47, Sh = 2.2$
	Occupancy	Hour	–	1	–	5	–	
Occupants	Occupancy	Hour	–	1	–	5	–	

Note: G (Gamma), Log (Logistic), LN (Log-Normal), N (Normal), W (Weibull), L (Location), Sh (Shape), Sc (Scale).

free-standing sensors. Due to the strong correlations between indoor air temperature ($R^2=0.85$) and CO₂ measurements ($R^2=0.8$), AHU sensor data were excluded from this study to enhance numerical processing efficiency. Mean value of indoor air temperature is well-conditioned within setpoint ($>21^\circ\text{C}$). These data showed that average indoor conditions were generally well-controlled. However, PD% of environmental factors are varied widely, highest for perceived dissatisfaction rate with indoor air quality (PD_{IAQ}, 52% mean), followed by PD_{TE} and PD_{WB} (%). However, student dissatisfaction often occurs obviously, especially on PD_{AE} and PD_{TE} (%). This showed deeper analysis is required to understand how all PD% varied widely. Perhaps, indoor data together with outdoor data are necessary input for predicting PD%.

3.2. ML input variables pre-selection by statistical test

The visualization presented in Fig. 3 shows the relationships between various environmental variables (indoor and outdoor) and dissatisfaction level distributions (*Very Dissatisfied*, *Dissatisfied*, *Satisfied*, or *Very Satisfied*) with p-values for significance indicator. CO₂ concentration (indoor) has significant impact on dissatisfaction levels with IAQ as well as indoor temperature with thermal environment, respectively ($p = 0.000$). Lower dissatisfaction levels with IAQ were observed when CO₂ levels were below 800 ppm, and thermal environment dissatisfaction was lowest when indoor temperatures ranged from (21.5–22.5 °C). VOCs ($p = 0.015$) also showed a notable relationship with dissatisfaction levels for IAQ, suggesting potential discomfort linked to elevated VOC levels. For outdoor environment parameters, temperature ($p < 0.001$), relative humidity ($p = 0.005$), and solar radiation ($p = 0.000$) demonstrated significant associations with dissatisfaction levels in the thermal and visual environments. Occupancy hours significantly influenced dissatisfaction with IAQ ($p = 0.000$) and thermal comfort ($p = 0.002$) with longer occupancy hours correlating to increased dissatisfaction. Briefly, the visualized comparison in Fig. 3 suggests that better environmental conditions can increase student satisfaction by up to 20% compared to worse conditions. CO₂ concentration, VOCs, and humidity aspects were identified as key factors strongly influencing IAQ. Specific humidity and air enthalpy were found to significantly impact dissatisfaction with the thermal environment, highlighting their combined

influence. This highlights the impact of environmental variables on dissatisfaction levels across multi-domain environmental factors, inline if poor IEQ negatively impacts educational outcomes [4,5]. Prolonged exposure to poor IAQ and undesired temperatures may pose environmental discomfort and health effect.

Factors such as building parameters and occupancy hours consistently demonstrated statistical significance across all environmental domains, emphasizing the importance of building layout and usage patterns. For outdoor environmental variables such as, air temperature, specific humidity, and mean global radiation exhibited significant relationships with thermal and visual environmental perceptions. This indicates the occupants' preference for seasonal conditions that are not cold and rich in solar radiation. As summarized in Table 5, the selection of environmental variables as pre-selected inputs for the ML models are shown, providing a robust foundation for configuring input parameters in ML training models.

Table 6.

3.3. ML models development & explanation

Various ML algorithms were developed to predict the PD% of multi-domain environmental factors. The models (using MLR, DT, and RF algorithms) were trained and tested with specified train/test ratio in Section 2.3, and a hyperparameter tuning process was applied to optimize their performance as shown in Table 4. Performance of models were evaluated using R^2 and RMSE metrics to compare the accuracy of the best-performing model as defined in Eq. (3) and Eq. (4). An overview of the evaluation results is presented in Table 7.

The ML results in this study demonstrated varying accuracies, differences between training and testing results, and distinctions among the algorithms. The higher RMSE values and lower R^2 values in MLR compared to DT and RF algorithms highlighted its limited ability to capture non-linear interactions among input parameters as predictors. DT models showed notable improvements, but overfitting was evident, as they achieved good training results with a significant gap in poorer testing results. RF, as an ensemble ML method, addressed the overfitting limitations of DT models RF models proved reliable for predicting perceptions of environmental factors, demonstrating robustness in handling

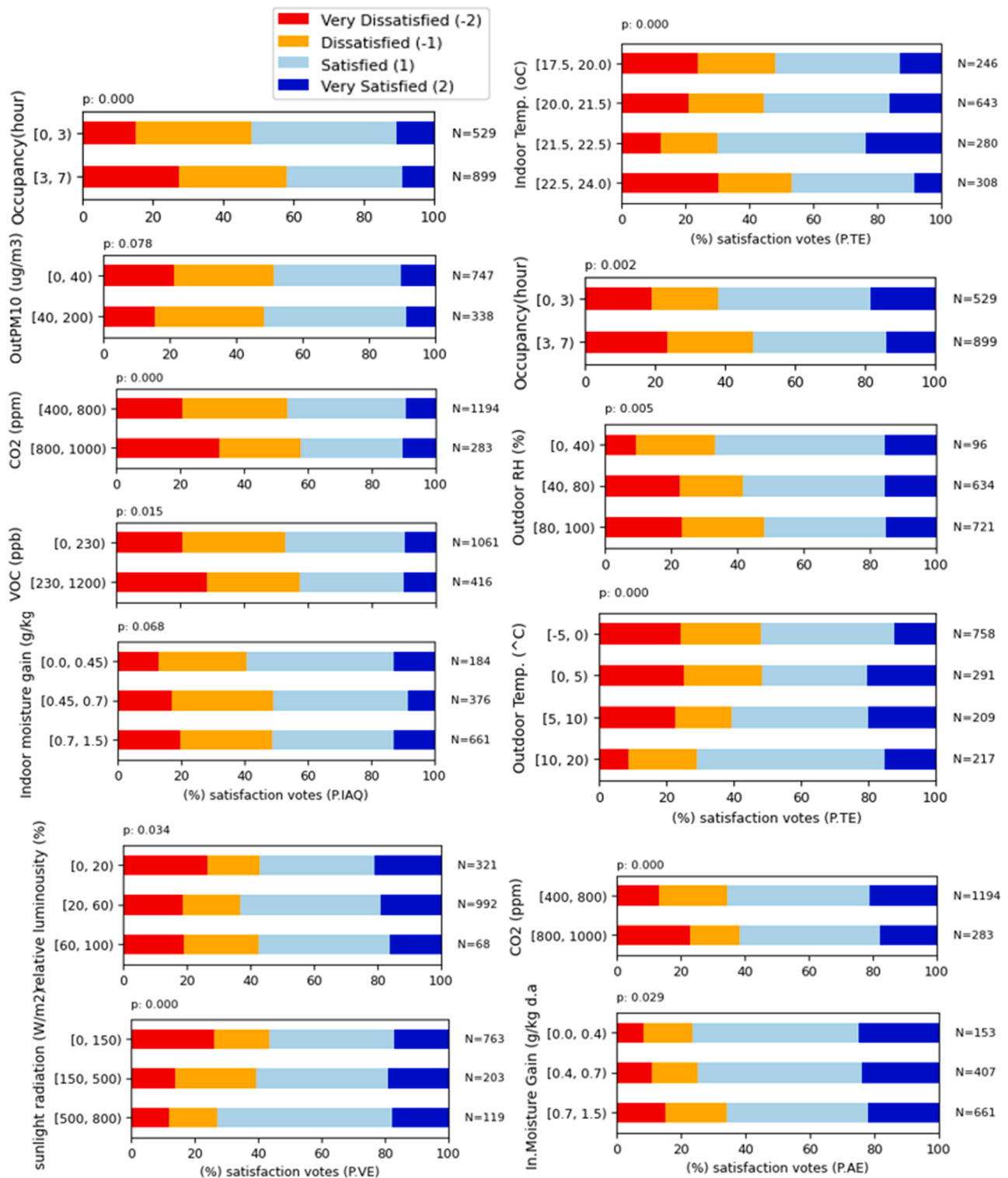


Fig. 3. relationship between various segmented environmental measurement results and student satisfaction-dissatisfaction votes distribution.

multi-variate input parameters. These results reinforced the opportunity of MLs use to predict PDs% by various environmental factors. Higher accuracy underscored the capability of RF algorithms to capture complex relationships between PD% and input parameters. RF algorithms showed notable results for PD_{TE} , PD_{VE} and PD_{IAQ} , achieving R^2 values of 0.74 and 0.86 on test datasets, respectively. RF model also performed better with R^2 0.74 on test data while predicting PD_{VE} but performed lower by applying MLR and DT algorithms. ML model in predicting PD_{AE}

achieved an R^2 of 0.86 (train) and 0.53 (test) with RF algorithm, indicating lower generalization capabilities.

Understanding how ML models work is critical for analysing the interaction between indoor conditions and PD% as the prediction output. Relative feature importance derived from SHAP analysis is presented in Fig. 4, visualizing the contribution of environmental variables to predict PD% of multi-domain environmental factors using RF algorithms in Table 5 as best-performing models. All input parameters

Table 6

Summary of statistical hypothesis test of students' dissatisfaction in relation to various environmental variables considered in ML input parameter selection by ANOVA & Chi-squared test contingency in Fig. 3.

Parameters:		Significance of Hypothesis Tests by Environmental Factors:			
		IAQ	Thermal	Visual	Acoustic
indoor env.	Carbon dioxide	a5, c1 *		a5	a5, c1 *
	Air Temperature		a1, c1 *		
	Relative Humidity		c1		
	Air Enthalpy	c1	a1		
	Specific Humidity	a1, c1 *	a1, c1 *		
	VOCs	a1, c1 *			a5
	Relative Light Intensity			a5, c5 *	
	Barometric Pressure	a5	a1		a5
	moisture gain	a5	a5		a1, c5 *
	Air Temperature	a1, c5 *	a1, c1 *		
outdoor env.	Relative Humidity				a5
	Wind Speed	a5			
	Precipitation	c5			
	Air Enthalpy	c1	a1		
	Specific Humidity	a1, c1 *	a5		
	Barometric Pressure		a1		a5
	Mean Global Radiation			a5, c1 *	
	PM10	a1	a1	a1	a1
	Room Floor location	a5, c1 *	a5, c1 *		a1, c5 *
	Façade area – west	a5, c1 *	a5, c1 *	a5, c1 *	a5, c1 *
building	Façade area – east	a5, c1 *	a5, c1 *	a5, c1 *	a5, c1 *
	occupancy time (hours)	a5, c1 *	a5, c1 *	a5, c1 *	a5, c1 *

Notes:

a1: ANOVA one-way tested – ($p < 0.01$).

a5: ANOVA one-way tested – ($p < 0.05$).

c1: Chi-squared tested – ($p < 0.01$).

c5: Chi-squared tested – ($p < 0.05$).

* : Parameters selected as the multi-input configuration to predict PD% for each environmental factor.

contributed differently to the predictions, as shown in their relative importance. Interaction between environmental conditions and building parameters may be happened. Indoor temperature and window area (west-façade) the most substantial influence, suggesting that heat gain from solar radiation through windows at west-façade significantly affects PD_{TE}. Indoor CO₂ levels, VOC levels, and indoor specific humidity showed significant relative importance for PD_{IAQ}%. Minorly, PD_{IAQ}% is supported by lesser extent by building parameters and also outdoor temperature as proximity of outdoor season. Light intensity was the dominant parameter for predicting PD_{VE}%, followed by outdoor global

solar radiation and supported by windows area at west-façade. PD_{VE}% indicates the needs of natural lighting through windows. The prediction of PD_{VE}% could be more comprehensive, if influencing factors such as window curtains, artificial lighting load, and gender [59] are included. However, these elements are often difficult to monitor digitally, posing a challenge for practical and privacy matters. This study also has lack of capabilities in understanding perceived acoustical environments (PD_{AE}%), ML with RF algorithm in this study understood CO₂ levels and indoor moisture gain as contributed inputs. Perhaps, there is contribution between human noise activity with indoor moisture gain. Even though CO₂ on humans co-influenced together with noise [60,61], CO₂ levels in this study poorly understands perceived acoustic in classroom. However, credibility of MLs may be optimized if indoor measurements can be supported by other relevant such as: sound pressure [34,38], PM levels [18,19], etc.,.

The results indicated that a multi-variable input configuration is essential for more accurate predictions. Dissatisfaction levels related to environmental factors also require consideration of outdoor environmental variables. This highlights the opportunity to improve the

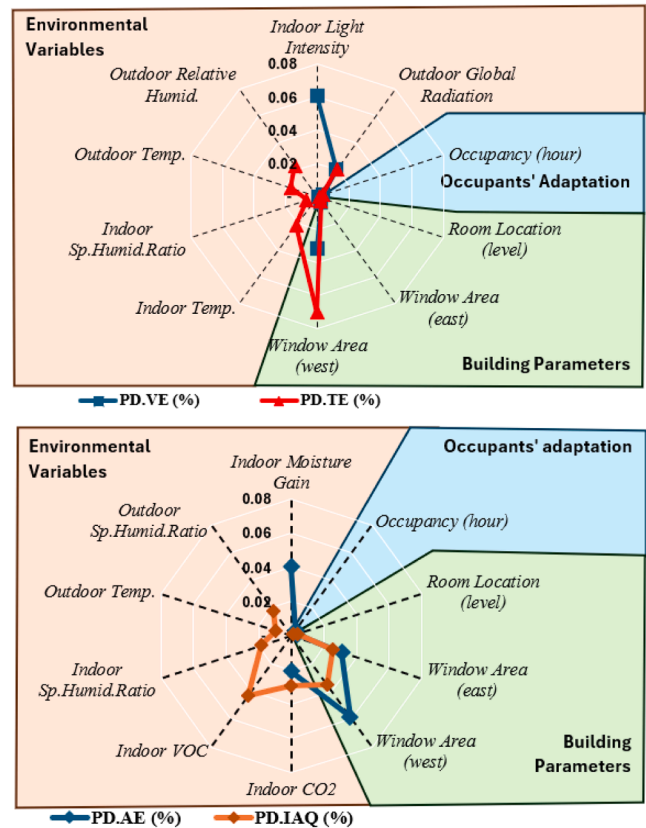


Fig. 4. Relative feature importance by SHAP analysis in observing impact of input variables.

Table 7

MLs prediction performance metrics by various environmental factors.

ML Algorithms	ACCURACIES OF PREDICTION:								Metrics
	PD _{IAQ}		PD _{VE}		PD _{TE}		PD _{AE}		
	Test	Train	Test	Train	Test	Train	Test	Train	
Linear (MLR)	0.23	0.36	0.67			0.70	0.23	0.45	R ²
Decision Tree (DT)	0.24	0.78	0.48	0.99	0.64	1.00	0.59	1.00	
Random Forest (RF)	0.74	0.81	0.80	0.87	0.85	0.91	0.63	0.85	RMSE
Linear (MLR)	0.13	0.16	0.08	0.10	0.15	0.12	0.09	0.14	
Decision Tree (DT)	0.15	0.09	0.10	0.02	0.14	0.05	0.20	0.03	
Random Forest (RF)	0.09	0.09	0.06	0.05	0.08	0.09	0.14	0.08	

prediction method for PD% of multi-domain environmental factors. ML models trained at the building level are designed to incorporate building parameters as critical components, addressing aspects that cannot be fully captured or measured using sensor devices. These include the unique conditions of individual rooms, pollutant and thermal gradients indoors, varying room views, visible sunlight through windows, occupant behaviours, walls and windows quality, acoustic performance, and student preferences across different spaces. Building parameters, such as windows area and floor location, not only reflect physical aspects like window size and the influence of floor differences but also represent unique room characteristics that cannot be measured by instruments. This remarks the importance of building parameters affecting occupants' satisfaction [20,21]. However, occupant adaptation, while being the most subjective factor and closest to the occupants in this study, played a less prominent role in the ML predictions. This may be because building and environmental variables were more dominant in determining dissatisfaction levels. Additionally, students moving in and out of classrooms frequently might have reduced their sensitivity to environmental changes, affecting the consistency of their assessments. Over the time, students may have found staying indoors increasingly uncomfortable, as reflected by the rising PD% values shown in Fig. 3.

3.4. Multi-domain environmental effect on students' well-being

Examining how environmental factors contribute to maintaining students' well-being in schools is essential, as these factors are significantly influenced by IEQ conditions [11]. A common approach to IEQ evaluation involves assessing perceived multi-domain environmental factors, with a primary focus on thermal, acoustic, and visual environments, as well as IAQ [6,10]. A linear equation comprising variables representing environmental factors is often used to aggregate IEQ components [62,63]. In this study, students' well-being is assumed to correspond to overall IEQ perception. Actual values of dissatisfaction rates for IAQ ($PD_{IAQ}\%$), thermal comfort ($PD_{TE}\%$), visual environment ($PD_{VE}\%$), and acoustic environment ($PD_{AE}\%$) are regressed to approximate well-being perception ($PD_{WB}\%$) as assumed as overall IEQ dissatisfaction ($PD_{IEQ}\%$) with null the intercept, as derived in Eq. (5). The analysis reveals that students' well-being is primarily influenced by IAQ, acoustic, and thermal environmental factors, with visual environmental factors only providing minor contributions as indicated by weight coefficients individually in Eq. (5).

$$PD_{IEQ} (\%) = 0.30 (PD_{IAQ}\%) + 0.23 (PD_{TE}\%) + 0.12 (PD_{VE}\%) + 0.35 (PD_{AE}\%)$$

$$(R^2 = 0.56; p < 0.01)$$

Linear equation in Eq. (5) may exhibit limited explanatory power

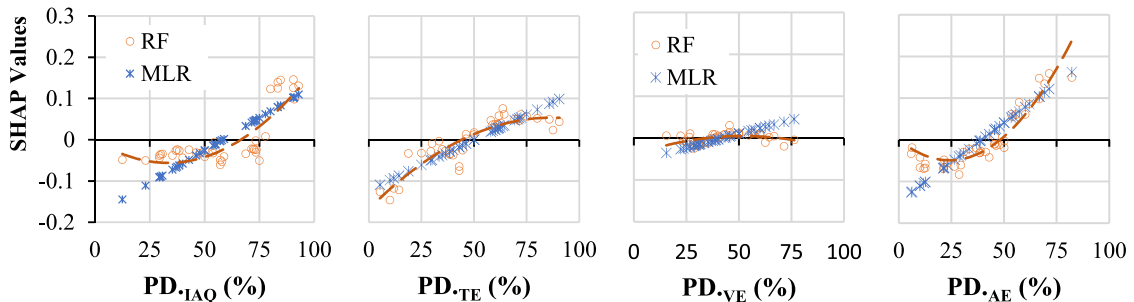


Fig. 5. Dependence plot of SHAP values while predicting well-being dissatisfaction rate by various real values of dissatisfaction rates (PD%) at multi-domain environmental factors: ML-MLR (R^2 : 0.56) vs. ML-RF (R^2 : 0.91).

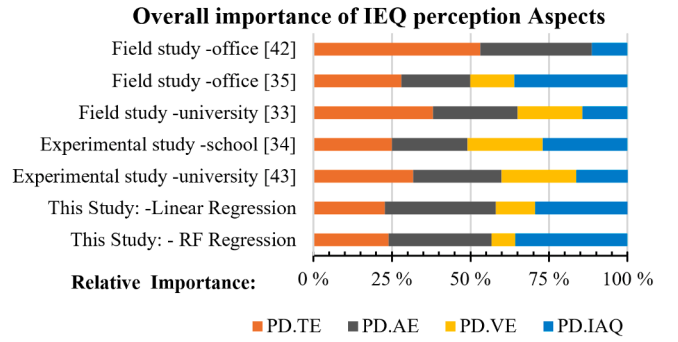


Fig. 6. Importance of environmental factors regarding the impact on overall satisfaction.

(R^2 : 0.56) due to its simplicity in calculating the proportional influence of each environmental factor on PD%. In contrast, ML modelling using the RF algorithm demonstrates promising accuracy in predicting $PD_{IEQ}\%$, achieving R^2 of 0.91. Scattered values of SHAP analysis further highlight the contribution of multi-domain environmental factors to $PD_{IEQ}\%$. The RF algorithm achieved better alignment between SHAP values of $PD_{IEQ}\%$ and actual PD% of environmental factors, compared to MLR, reveals the non-linear connections between the PD% of environmental factors and $PD_{IEQ}\%$, as shown in Fig. 5. Moreover, RF proved more effective in predicting $PD_{IEQ}\%$, whereas MLR exhibited inaccurate prediction, resulting in lower accuracy (R^2 values - RF: 0.91 vs. MLR: 0.56). As shown in Fig. 6, the findings of this study slightly align with four previous studies evaluating environmental factors [34–38]. By comparing the regression coefficients or parameter vectors derived from MLR in Eq. (5) with the relative importance obtained from the trained ML algorithm (RF) in Fig. 5, both models consistently assigned higher importance to IAQ, acoustic, and thermal environmental factors. This emphasizes the priority of addressing these factors to improve IEQ in classrooms.

ML offers practical solution for implementing indoor environment optimization strategies in school environments, ensuring healthier conditions for students. By proposing these ML methods, this study introduces the concept of satisfaction level prediction, enabling dynamic adjustments to maintain optimal indoor conditions. A key limitation of this study is its focus on a specific school with three classrooms in a cold-

climate region, involving students aged 13–15. This narrow scope may limit the generalizability of the findings to other occupants, buildings, and environmental conditions, as actual PD% of all environmental

factors may vary significantly. Relative feature importance values highlight the key contributing factors among input parameters for predicting PD%. Despite its limitations, it is challenging, if not impossible, to develop a universally applicable IEQ equation model [64].

This study presented an ML development framework utilizing a multi-variate technique to predict PD% of environmental factors at the local building scale, providing a foundation for further improvements and investigations. As practical suggestion, implementation of ML to predict dissatisfaction is feasible to be deployed in building management system, enabling building operators to consider comfort strategies and optimize well-being. Moreover, practical strategies may apply to be done simultaneously. Clothing strategies encouraged by teacher to students with appropriate clothing, such as layering, can help students adapt to temperature variations [65]. Besides, environmental cleaning strategies reduces pollutive contaminant accumulation, further improving students' health [66]. Structuring student activity schedules to include breaks with clean environment allows exposure to fresh air, promoting better well-being [67,68]. By this one example of solution, indoor well-being might be optimised as IAQ, thermal and acoustic environment influence much to overall IEQ perception.

4. Practical implication and limitations

This study offers valuable insights for both research and practice in the field of IEQ, particularly within educational settings. By integrating student perception data with environmental sensor measurements and applying ML models, this work bridges the gap of IEQ perceptions between subjective feedback and environmental variables. The findings support future academic research and inform practical decision-making for school facility managers, policymakers, and designers of healthier learning environments. The practical implications of this study include:

- a) **New Approach:** This study introduces a predictive model for perceived dissatisfaction (PD%) across multiple environmental domains—IAQ, thermal, acoustic, and visual. Traditional models often rely on fixed thresholds or standardized comfort equations (e. g., PMV-PPD for thermal comfort), which may not reflect actual occupant experiences. The ML approach presented here enables localized predictions based on real occupant feedback, which is particularly important in schools, where students' perceptions and sensitivities can vary significantly. Generalized formulas—such as universal thermal comfort models [28]—may not be suitable in this context due to local-specific conditions. Therefore, localized modeling provides more accurate and relevant support for decision-making.
- b) **Specific Applications and Operational Use:** The integration of repeated student feedback allows for the development of predictive tools tailored to specific environmental variables and room characteristics. Seasonal recalibration may also be applied, enabling the model to adapt to varying comfort preferences across different occupants and times of the year [69,70]. This local modelling approach makes it feasible for school authorities to incorporate such frameworks into Building Management Systems (BMS) for real-time monitoring. The application of ML algorithms under data analysis methods offers a new utilization opportunity for continuous monitoring and operational insight in IEQ systems, particularly within educational settings. ML models enable schools to move beyond static threshold monitoring toward dynamic, occupants' centred management strategies. In doing so, ML-supported IEQ frameworks can facilitate proactive interventions that better align environmental control with students' actual comfort needs. This study also highlights the challenge of implementing ML models based on the available measurement instruments in a building. However, if Building Management System (BMS) services are equipped with more comprehensive sensors, it would significantly enhance the value of ML applications. Overall, it opens opportunities for

integrating IEQ monitoring with ML-based predictive tools to improve the accuracy of overall IEQ and multi-domain environmental perceptions.

This study comes with specific uniformed condition (students age, location, general occupants' activity, building design, measurements availability and accuracy, etc.). In Norway, schools do not enforce a uniform policy; however, clothing was not systematically recorded or analysed as it is assumed as adjusted individually by students already wearing their comfortable set of clothes from home. Instrument and data availability also limited in implementing this ML method. In this case study as example, no sound pressure level sensors were available which limited PD_{AE}% prediction, and less monitoring data on artificial lighting, and windows opening aspect for improving PD_{VE}% prediction. Outdoor data may also present limitations. Primary data requires additional instruments for on-site outdoor measurements, while secondary sources, such as weather stations, may lack complete information on weather and pollution variables in certain areas.

Future research may adopt this approach across broader specification, enabling the optimization of ML models through hyperparameter tuning and cross validation, ultimately contributing to a deeper understanding of perceived IEQ and the development of tailored ML-based solutions. Incorporating additional environmental variables, such as particulate matter or sound pressure levels, may further enhance the predictive accuracy of the models.

5. Conclusions

This study aimed to predict students' perceived dissatisfaction rates (PD%) across multiple indoor environmental quality (IEQ) domains—IAQ, thermal, acoustic, and visual—within a school setting using machine learning (ML) models. By integrating building parameters, indoor and outdoor environmental measurements, and 1437 student survey responses collected longitudinally, the study revealed that high PD% values can still occur even when environmental conditions appear within normal or controlled ranges.

Critical input variables included indoor CO₂, VOCs, temperature, humidity, and light intensity, with building characteristics—such as room orientation and window area—also contributing significantly to PD% predictions. Outdoor weather variables, particularly temperature, humidity, and solar radiation, influenced thermal and visual dissatisfaction, highlighting the importance of considering both indoor and outdoor factors.

The findings demonstrate that ML models, particularly Random Forest (RF), can reliably predict PD% across environmental domains using relevant multi-input configurations. RF outperformed Multi-Linear Regression (MLR) and Decision Tree (DT) models, achieving strong testing accuracies for IAQ ($R^2 = 0.74$), thermal ($R^2 = 0.80$), and visual environments ($R^2 = 0.85$). Unlike MLR, which struggled to capture non-linear relationships, RF demonstrated better generalization with less overfitting. Non-linear ability of RF model optimized the highest accuracy in predicting overall IEQ dissatisfaction (PD_{IEQ}), with R^2 reaching 0.91, far surpassing MLR ($R^2 = 0.56$). The results consistently showed the ML algorithm holds strong potential for implementation, with acoustic, IAQ, and thermal environment perception domains are the most influential factors in shaping students' overall IEQ perceptions. These insights provide a strong foundation for developing data-driven monitoring and building-specific strategies for improving indoor environments in educational settings.

CRediT authorship contribution statement

Azimil Gani Alam: Writing – original draft, Visualization, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Alena Bartonova:** Writing – review & editing, Supervision, Resources, Project administration, Investigation, Data curation, Conceptualization. **Britt**

Ann Kåstad Høiskar: Supervision, Resources, Project administration, Investigation, Funding acquisition. **Mirjam F. Fredriksen:** Visualization, Validation, Software, Resources, Investigation, Data curation. **Jivitesh Sharma:** Writing – review & editing, Supervision, Software. **Hans Martin Mathisen:** Writing – review & editing, Supervision, Formal analysis, Conceptualization. **Zhirong Yang:** Writing – review & editing, Validation, Supervision. **Kai Gustavsen:** Supervision, Methodology, Formal analysis. **Kent Hart:** Supervision, Formal analysis. **Tore Fredriksen:** Supervision, Project administration, Funding acquisition. **Guangyu Cao:** Writing – review & editing, Supervision, Methodology, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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